



Review

Recent trends in advanced oxidation process-based degradation of erythromycin: Pollution status, eco-toxicity and degradation mechanism in aquatic ecosystems



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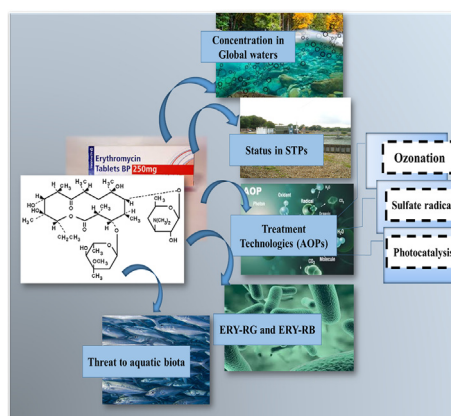
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HIGHLIGHTS

- Providing ERY concentration and entry evidences in global waters
- Highlighting the status on ERY in STPs
- Evidences regarding the rise of ERY-RG and ERY-RB in global water entities
- Underlining serious threats posed by ERY on aquatic biota
- Reactive radical production underlining O_3 , $SO_4^{\cdot-}$, $HO^{\cdot}$ and $O_2^{\cdot-}$ contributions for ERY degradation

GRAPHICAL ABSTRACT



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ABSTRACT

Wide spread documentation of antibiotic pollution is becoming a threat to aquatic environment. Erythromycin (ERY), a macrolide belonging antibiotic is at the top of this list with its concentrations ranging between ng/L to a few $\mu\text{g/L}$ in various global waterbodies giving rise to ERY-resistance genes (ERY-RGs) and ERY-resistance bacteria (ERY-RBs) posing serious threat to the aquatic organisms. ERY seems resistant to various conventional water treatments, remained intact and even increased in terms of mass loads after treatment. Enhanced oxidation potential, wide pH range, elevated selectivity, adaptability and greater efficiency makes advance oxidation processes (AOPs) top priority for degrading pollutants with aromatic rings and unsaturated bonds like ERY. In this manuscript, recent developments in AOPs for ERY degradation are reported along with the factors that affect the degradation mechanism. ERY, marked as a risk prioritized macrolide antibiotic by 2015 released European Union watch list, most probably due to its protein inhibition capability considered third most widely used antibiotic. The current review provides a complete ERY overview including the environmental entry sources, concentration in global waters, ERY status in STPs, as well as factors affecting their functionality. Along with that this study presents complete outlook regarding ERY-RGs and provides an in depth detail regarding ERY's potential threats to aquatic biota. This study helps in figuring out the best possible strategy to tackle antibiotic pollution keeping ERY as a model antibiotic because of extreme toxicity records.

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1. Introduction

Wide use plus abuse of antibiotics for human as well as veterinary disease treatment has now involved public distresses (Barbara Ambrosetti et al., 2015; Chen and Zhang, 2013; Chen et al., 2020; Richardson et al., 2008; Rodríguez-mozaz et al., 2014; Wang et al., 2019). Serious pollution cases are reported due to inefficient antibiotic handling leading to an increased concentration especially in water bodies (Göbel et al., 2007; Wang and Zhuan, 2020). Due to their non-biodegradable nature they provide a breeding ground to antibiotic resistance bacteria (ARB) resulting in antibiotic resistance genes (ARGs) formation (Chen et al., 2013; Chen and Zhang, 2013; Davies and Davies, 2010; Di et al., 2016; Gao et al., 2012; X. Ji et al., 2012; Ling et al., 2013; Martínez, 2008). Out of all other globally used and misused antibiotics a macrolide belonging antibiotic ERY is considered as the priority due to its excessive usage.

In the decision 2015/495/EU monitored by European Union in a watch list released in 2015 ERY was included as a high-risk substance. The rank list of different pollutants developed in China on the basis of their presence in the treated water or waste water treatment plant effluents ERY has been labelled as one of the most treatment plant effluents ERY has been labelled as one of the most treatment plant effluents ERY has been labelled as one of the most treatment plant effluents

pollutants, arduous to be degraded naturally (Islas-Espinoza et al., 2018). Additionally, ERY binds with ribosomal RNA hindering protein synthesis (Pérez et al., 2017). It is a two sugar containing polyhydroxylactone. Erythronolide portion contains 14 membered lactone ring. D-desosamine amino sugar is attached to the C-5 portion of the lactone ring. The alkaline nature of ERY is due to this attached desosamine. The other sugar called L cladinose is unique to ERY that is attached to C-3 portion of the lactone ring firmly through a β-glycosidic bond (Kanfer et al., 1998; Lange et al., 2006; Roberts, 2014) as shown in Fig. 1. Number of studies reported the presence of ERY traces in various water bodies including rivers (Gao et al., 2012a; Jiang et al., 2014; Wang et al., 2017; Zhou et al., 2011), lakes (Ding et al., 2017; Liu et al., 2018) and in waste water effluents (Gao et al., 2012b; Göbel et al., 2007; Karthikeyan and Meyer, 2006; Miao et al., 2004). Along with that, many studies reported toxic levels of ERY in various organisms. Aquatic photosynthetic organisms prominently cyanobacteria are affected at approximated ERY concentration of 22–350 µg/L prohibiting its growth and metabolism (Gonzalo et al., 2013). Moreover, highlighted liver damage was recorded in fish (Liu et al., 2014; Rodrigues et al., 2018).

ERY removal from the waste water has been investigated in various studies. Conventional techniques fail due to elevated operational cost

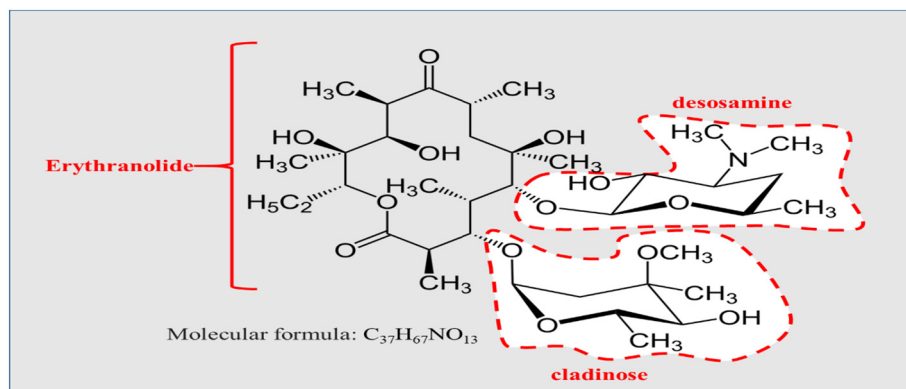


Fig. 1. Structure of ERY.

and declined degrading efficiency. Contrary to them advanced oxidation processes (AOPs) showed positive response in tackling the antibiotic pollution. In case of ERY degradation various AOPs seem very effective as highly strong oxidative agents such as hydroxyl radical ($\text{OH}\cdot$) (Xekoukoulotakis et al., 2010) in photocatalytic oxidation, ozone (O_3) (Lange et al., 2006; Tay et al., 2010) in Ozonation, superoxide radical ($\text{O}_2^{\cdot-}$) (Chu et al., 2019; Hua et al., 2019; Teel et al., 2016) and active persulfate (PS) used in these processes enhances the antibiotic degradability converting it into small molecule substances.

The purpose of this review is to provide extensive sources, existence and eco-toxic impacts of ERY providing detailed picture regarding fate, behavior and concentration of ERY in global waters highlighting the emerging threat of ERY-resistance genes (RGs) and ERY-resistance bacteria (RBs) as well as unfolding ERY status in STPs along with emphasizing its degradation efficiency through various AOPs including, ozonation, sulfate radical ($\text{SO}_4^{\cdot-}$) based AOPs and photocatalysis with their brief knowhow, principles, basic characteristics, major influencing factors along with their ERY degradation potential in water and wastewater.

2. Methodology

Information related to the ERY treatment technologies, mechanism, degradation efficiency and evidences, was obtained by searching literature and article from peer-reviewed journals using the scientific databases and various search engines including Web of Science, Google Scholar, Science Direct, NCBI and other major publishers. The rummages were accomplished using blends of the following keywords: ERY degradation, sewage treatment plants and ERY; Ozonation for ERY; Advanced Oxidation Techniques; photocatalysis and ERY; and ERY and global waters, eco-toxicity and ERY and bioaccumulation of ERY. A total of 178 articles were considered to be reviewed for this manuscript. Major literature was opted from reputed journals and various book sources.

3. Fate, behavior and source of ERY in the environment

McGuire and co-workers discovered ERY initially produced in soil system (Luiz et al., 2010) in 1952 as metabolic product of *Streptomyces erythraeus* strain, in future allotted to the genus *Saccharopolyspora*. ERY

can be produced naturally as these bacteria inhabit in soil. To produce ERY at commercial scale *Saccharopolyspora erythraea* has been used. Production of ERY by the above mentioned bacterium enabled the subsistence towards harmful organisms that resided in the soil environment. The considerable portion of this antibiotic enters in soil through anthropogenic activities which ultimately found in groundwater (Schafhauser et al., 2018). Studies witnessed its vulnerability to direct photolysis as it absorbs light at wavelength > 290 nm. In water environment its combination with various sediments has been witnessed. Volatilization is not possible as its pKa is 8.9 and exists in the environment in cationic form (Roberts, 2014). Frequent detection of main ERY metabolite i.e. ERY- H_2O is observed in various studies. In acidic conditions loss of one molecule of water because of instability transforms ERY into its metabolite (Blaney, 2014; Hirsch et al., 1999). The primary ERY entry sources in the environment are drug disposal most probably from pharmaceutical industries as well as households (Qarni et al., 2016). Human and animal excretions are other major entry sources (Høverstad et al., 1986; Sarmah et al., 2006). A study reported the concentration of ERY in humans excreted urine between 0.02 and 20% (Pérez et al., 2017). Another major ERY environment entry source includes the water and sewage treatment plants (Acuña et al., 2014; Göbel et al., 2007). At different levels of treatment ERY was found to be quite resistant (Leung et al., 2012). In water treatment procedure ERY combines with the bio solids contaminating the sludge, that sludge when further applied as a fertilizer contaminate the soil which ultimately transport ERY residue to the ground water or to the nearest surface water bodies through runoff (Yan et al., 2014). Veterinary waste is also a highlighted ERY environment introduction source (Chen et al., 2020) and manure can act as ERY transporting tool (Bartelt-Hunt et al., 2011). Aquaculture effluents have been reported and categorized as ERY sources (Bottoni et al., 2010). The detailed entry sources of ERY in the environment are shown in Fig. 2.

4. ERY pollution in global waters

ERY concentration was highlighted mostly in urban waterbodies, as waste water treatment plant effluent discharge represents a major antibiotic source in the environment. This is due to discharge of waste water from pharmaceutical industries, hospitals, and domestic sources into the

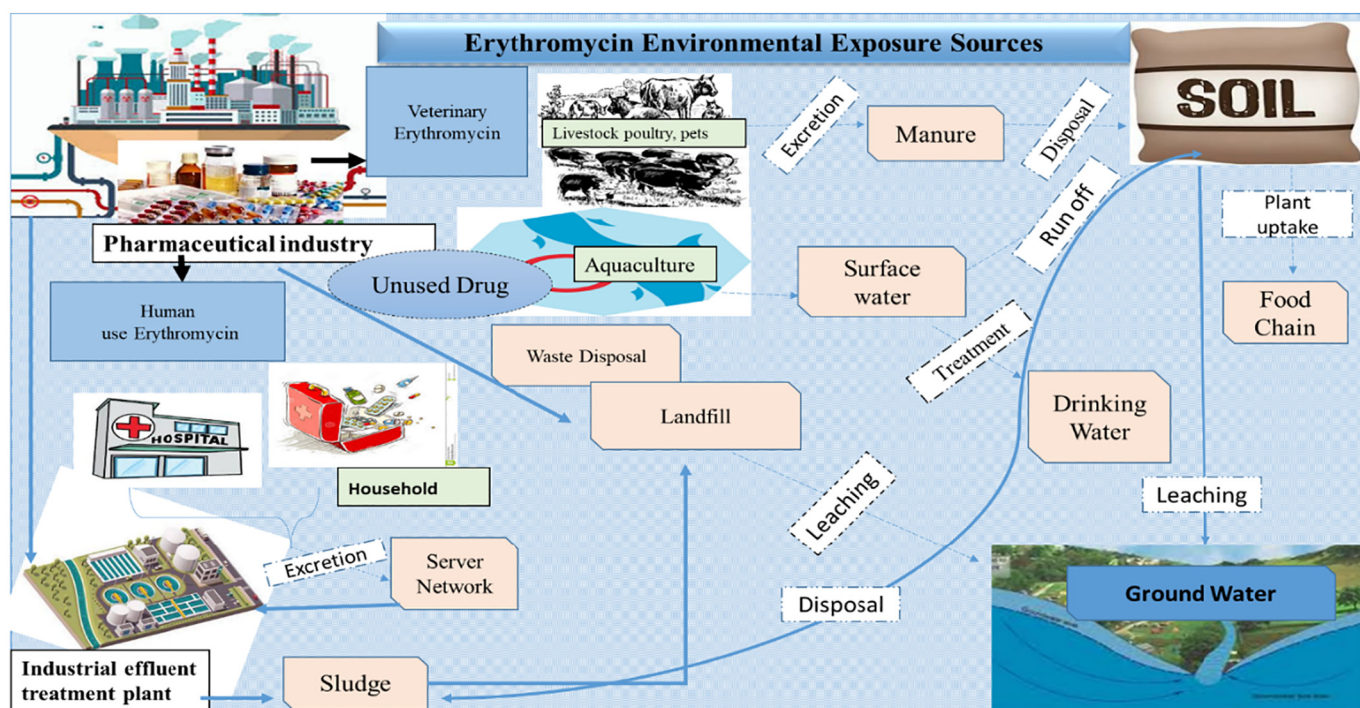


Fig. 2. Pathways of ERY in the environment.

freshwater system (Zheng et al., 2012). For instance, Taihu Lake is the third largest freshwater lake in China. The lake is surrounded by highly developed cities and towns, which account for >14% of gross domestic product of China. Numerous wastes from industrial and agricultural activities and municipal sewage from small towns enter the lake, causing severe environmental problems, such as antibiotic pollution. ERY concentration recorded in this river was 218.3 ng/L (Xu et al., 2014).

Similarly, the concentration of ERY in global river systems was keenly studied 8 to 24 ng/L in Yangtze river (Chang et al., 2010), 40 to 560 ng/L in Pearl River, China (Peng et al., 2011), 80.3 to 160.5 ng/L in Ume River, 347.4 to 694.8 ng/L in Dal River Sweden (Lindim et al., 2016). In Taiwan the maximum ERY-H₂O concentration was 76 µg/L (Minh et al., 2009), 30–7 ng/L in Elbe river Germany (Wiegel et al., 2004). W. hai Xu et al. (2007) stated the concentration ERY in Pearl river China ranges from 30 to 460 ng/L, 2806 ng/L Labenon Rivers (Mokh et al., 2017).

Further the presence of ERY in different water treatment plants was analyzed, 139 stream sites in US were reportedly vulnerable to contamination with 21.5% ERY-H₂O frequent detection. Average ERY-H₂O concentration was reported to be 1.7 µg/L (Kolpin and Meyer, 2002). A study reported the highest ERY concentration of 10,025 ng/L in an underdone sewage (Kasprzyk-Hordern et al., 2009). ERY contamination in groundwater grabbed increasing attention, as ground water is used for drinking purposes globally. A study reported ERY concentration 2380 ng/L in groundwater (Bartelt-Hunt et al., 2011). In another study the wells in Baden-Württemberg, Germany were analyzed reporting the presence of 8 antibiotics including ERY-H₂O in groundwater out of 60 (Sacher et al., 2001), 0.049 µg/L ERY was reported in another study carried out in groundwater of Germany (Miao et al., 2004). Further detail is provided in Table 1.

5. ERY status in sewage treatment plants (STPs)

STPs are considered gateways for ERY entry in the environment (Gao et al., 2012b; Thomsen et al., 2005). Various studies report incomplete

degradation of antibiotics in treatment plants the reason is the absence of complex mechanism required for antibiotic degradation (Leung et al., 2012). Different STPs eliminate contaminants at different levels as the elimination rates depend on factors including the hydraulic retention time (HRT), the opted sort of treatment, solids retention time (SRT), even rainwater input along with temperature. Thus, the rate of contaminant elimination varies plant to plant and sometimes in a single plant it differs at multiple time periods (W. Xu et al., 2007). ERY removal efficiency is calculated by influent analyte concentration (C_{in}) and effluent analyte concentration (C_{ef}).

$$\text{ERY removal efficiency} = [(C_{in} - C_{ef}) / C_{in}] \times 100\%.$$

The antibiotic removal efficiency of various treatment mechanisms including activated sludge system, fixed bed and membrane reactors need a further analysis (Göbel et al., 2007). Various treatment mechanisms are shown in Fig. 3.

The removal efficiency of macrolides is quite low as equated to other antibiotics. Degradation proficiency of ERY ranges from –74 to 73%. One more study reported negative ERY removal efficiency by STPs (Richardson et al., 2008). The possible reason described in different studies was the exclusion of particles >0.45 µm for analysis leading to concentration underestimation in the influent (Petrovi et al., 2005). Similarly, during the treatment process, alteration in physiochemical parameters due to changing analyte behavior including adsorption can highly effect the ERY removal efficiency. Moreover, de-conjugation of the conjugated metabolites in the raw influent can be a reason to it (Gao et al., 2012b; W. Xu et al., 2007). W. Xu et al. (2007) reported that during the sewage treatment process ERY-H₂O was stable and further investigations confirmed the increase of ERY-H₂O concentration in the final effluents. The study reported 26% removal efficiency of ERY-H₂O for that four different STPs in Pearl River Delta. The highest ERY-H₂O concentration recorded was 1978 ng/L in the influent. The reason

Table 1
ERY occurrence in world waters.

Location	Water bodies	Ranges (ng/ L)	References
China	Yangtze River	8 to 24	(Chang et al., 2010)
	Pearl River	40 to 560	(Peng et al., 2011)
		385–0.97	(Taylor et al., 2011)
	Baiyangdian Lake	to 155	(Li et al., 2012)
	Taihu Lake	218.3	(Xu et al., 2014)
		6.7	(Wu et al., 2014)
	Haihe River	12	(Zhou et al., 2011)
	Songhua Jiang River	0.2 to 11.5	(Wang et al., 2017)
	Hai River System	4200	(Chen et al., 2018)
	Huangpu River	0.4–6.9	(Chen and Zhou, 2014)
	Wangyang River	253.1	(Jiang et al., 2014)
	Beijing River System	372	(Li et al., 2015)
	Haihe River	45.1	(Gao et al., 2012a)
	Poyang Lake	10.7	(Ding et al., 2017)
	Chao lake	20.7	(Wu et al., 2014)
Sweden	Ume River	80.3 to 160.5	(Lindim et al., 2016)
	Dal River	347.4 to 694.8	
South Korea	Han River, S	3 to 14	(Choi et al., 2008)
Labenon	Labenon Rivers	2806	(Mokh et al., 2017)
Iran	Kan River	18.02	(Mirzaei et al., 2019)
	Firozabad River	45.43	
Ghana	Rivers of Ghana	1149	(Azanu et al., 2018)
Germany	German Rivers	1700	(Hirsch et al., 1999)
	Elbe river	30–7	(Wiegel et al., 2004)
US	Cache La Poudre River	120	(Kim and Carlson, 2007)
France	Predecelle River	4.2	(Tuc et al., 2011)
	Charmoise River	131	
UK	Taff River	121	(Kasprzyk-Hordern et al., 2009)
	Ely River	72	
Vietnam	Mekong River	9–41	(Satoshi Managaki, Ayako Murata, Hideshige Takada, Bui Cach Tuyen, 2007)
Beibu Gulf	Beibu Gulf	1.1–50.9	(Zheng et al., 2012)
Bohai Bay	Bohai Bay	150	(Zhou et al., 2011)

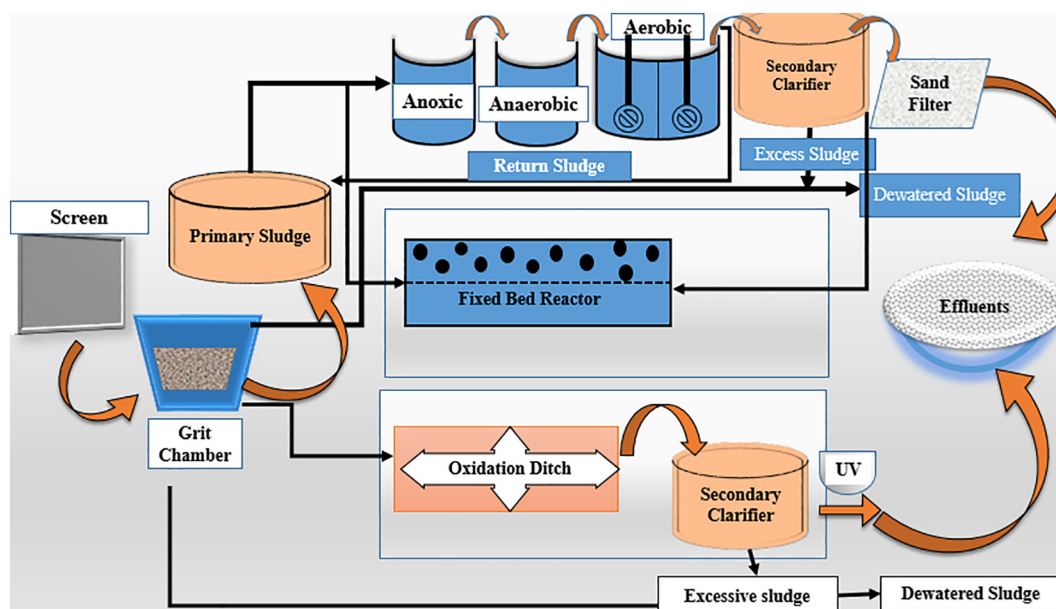


Fig. 3. Schematic diagram of various treatment methods observed in STPs (Zhou et al., 2013).

behind the high concentration of compound ERY-H₂O was its high consumption in the study zone. ERY-H₂O was found to be the dehydrated compound of ERY previously. A study recorded maximum ERY-H₂O in raw influents 0.19 µg/L. The elimination rate by means of conventional activated sludge treatment was recorded as –22 to 6% (Göbel et al., 2007). Similarly, a study in Spain recorded 2.31 µg/L elimination rate (Perdigo et al., 2010) where as ERY elimination through conventional treatment plant in Canada was recorded as 838 µg/L (Miao et al., 2004). In UK STPs Coslech and Cilfynydd ERY-H₂O was found at exceedingly high level i.e. 4 mg/L in raw sewage. The nature of treatment mechanism was responsible for the removal of ERY, as in STP Coslech waste-water treatment eliminated 50% of ERY-H₂O and in STP Cilfynydd no elimination of these antibiotics was witnessed. The reason behind was the use of activated sludge in STP Coslech and trickling filter beds in STP Cilfynydd as these are two different biological procedures used in two wastewater plant (Kasprzyk-Hordern et al., 2009). For compounds like ERY-H₂O sorption to activated sewage sludge was found to be less important, and probably sorption constants (K_d) ranges between 114 and 460 L/kg. On an average 200 g/m³ of specific sludge production eliminates compound with K_{db} 500 L/kg by <10% through sorption on to activated sludge (Thomsen et al., 2005). Detailed status of ERY in various STPs is shown in Table 2.

6. ERY resistance genes

ERY's extensive use in agriculture industry over the last half century is due to factors including its low cost, decreased mortality, livestock growth along with fertility benefits. Most specifically a steadily increased use of ERY, as a food additive has been observed in swine and poultry and is used for medicinal purposes (Knapp et al., 2010). The ERY-resistant bacteria (RB) strain selection could be expedited by an immense amount of ERY release in the environment (Roberts, 2014).

Rise in the use of recycled water in agricultural practices, agricultural runoff and waste water discharges whether treated or untreated, might have one to one concern for autochthonous bacterial colonies, particularly in limnetic ecosystems causing harmful environmental impacts (Proia et al., 2016; Rodríguez-mozaz et al., 2014). Ubiquitously, it has been realized that number of antibiotics belonging to various class are testified in several water entities, including the presence of ERY in a pointedly high concentration around the sphere (Guo et al., 2015; Singh et al., 2019). Transfer through food chain make it an eminent

threat to public health as ARGs are shifted through horizontal gene transfer in distantly related organisms and even between pathogens and non-pathogens (Lekunberri et al., 2017; Pazda et al., 2019).

ARG occurrence in fact is a serious emerging threat to aquatic environment (Baquero et al., 2008; Martinez, 2009; Munir et al., 2010; Zhang et al., 2016). Different ARGs are associated with different antibiotics. They have a potential to persist in case of antibiotic removal or inactivation of host bacteria and are observed to be more harmful than the antibiotic itself as they can transfer and migrate in various medias, carrying and long environmental persistence time. Their location is spotted on various mobile genetic elements. These elements include transposons, plasmids and integrons. Through horizontal gene transfer method they migrate between various species that result in the generation and wide spread of ARBs, posing serious human health risks (Lekunberri et al., 2017; Pazda et al., 2019). Therefore, they are categorized as one of the most significant 21 century challenges by WHO (Shen et al., 2019).

The proliferation hubs of ARB and ARG are wastewaters. Inefficient treatment of wastewater leads to the expansion of such problems (Guo et al., 2015; Lu et al., 2019; Santos et al., 2013). The supreme mode ARGs buildup is lake and river (Proia et al., 2016; Yang et al., 2018). ARGs and ARBs are abundant in limnetic environment, in China (Chen et al., 2015, 2013; Chen and Zhang, 2013; Ling et al., 2013). It has been recorded through various evidences by detection of various rivers across the world that there is an increase in the spillage of antibiotics has been increasing continuously (Liu et al., 2018; Lu et al., 2019). Bacteria when exposed to high concentrations, as a counter response they develop resistant elements and this is an obvious reason behind the presence of ARBs and ARGs in the water bodies across the world. Tello et al. (2012) highlighted that even a very low concentration of persistent ERY exposure in the waste water system persuaded a pressure promoting antibiotic resistance development. With exposition to ERY bacteria develop certain genes that can create resistance towards this particular antibiotic requiring a more powerful antibiotic for the next use, globally trapping individuals in a vicious cycle. The persistent exposure to ERY in wastewater, even at a low concentration, might induce a selective pressure for the resistant bacteria and promote the evolution of antibiotic resistance (Gao et al., 2015). Fan and He (2011) proved the presence of several ERY-RGs in gram-positive/gram-negative, aerobic/anaerobic bacteria and remained frequently detected in wastewater. Significantly, mobile elements result in proliferation of some ERY-RGs

Table 2
ERY status in various STPs.

Studies	STP locations	ERY-H ₂ O (ng/L)		HR (h)	SR (h)	Removal Efficiency (%)	Treatment processes
		Influent	Effluent				
(Richardson et al., 2008)	Wan Chai	8107 ± 11	8507 ± 2.1	–	–	–5	Primary treatment screening plant that screens suspended matter with diameter 46 mm
	Tai Po	4707 ± 2.5	5207 ± 8.5	16	20	–12	Secondary treatment processes including (i) removal of coarse solids; (ii) removal of grit; (iii) primary sedimentation; (iv) biological treatment for primary sewage; and (v) anaerobic digestion for surplus activated sludge
	Shatin	7407 ± 14	6007 ± 11	21	20	19	Secondary treatment processes including (i) screening of coarse material; (ii) settlement of grit particles; (iii) primary sedimentation of suspended matter; and (iv) biological biological treatment of sewage
	Stonecutters Island Shenzhen	5507 ± 8.0	5107 ± 9.0	1–2	–	9	Treatment process Chemically enhanced primary treatment process in which chemicals, including ferric iron and polymers, are used to treat the sewage. Treated sewage is discharged via an outfall in the western part of Victoria Harbor Secondary
(W. Xu et al., 2007)	Kaifu/Guangzhou	7517 ± 109	4307 ± 73	22	4.5–6	–	Directed to the activated sludge system for denitrification and nitrification
	Liede/Guangzhou	19,787 ± 233	20,547 ± 386	12–18	–	26	Oxidation ditch
	New Territory/Hong Kong	2537 ± 22	2167 ± 34	15–22	5.6–8.2	–	Directed to the activated sludge system for denitrification and nitrification
	Kowloon/Hong Kong	4697 ± 38	627 ± 24	8–12	3–4	–	Chemically enhanced
(Kasprzyk-Hordern et al., 2009)	Cilfynydd	242–6755	292–2841	–	–	0%	Biological process, using trickling filter beds, combining initial solids removal, biological secondary treatment and final clarification
	Coslech	144–10,025	23–2772	–	–	50%	Biological activated sludge process, deployed as an extended aeration oxidation ditch for BOD/COD removal and nitrification
(Gobel et al., 2005)	Kloten-Opfikon (WWTP-K)	–	–	30.1	–	–	Secondary treatment performed in two consecutive activated sludge unit's final effluent discharged to the receiving surface water after sand filtration as tertiary treatment
	Altenrhein (WWTP-A)	–	–	–	–	–	Secondary treatment is performed in two parallel-operated treatment units: a conventional activated sludge system and a fixed-bed reactor, receiving ~50% of the primary effluent each.
(Zhou et al., 2013)	China (Plant A)	179 ± 27.5	78.3 ± 45.5	–	–	28.9	Activated sludge treatment followed by chlorination
	China (Plant B)	–	–	–	–	77.9	Oxidation ditch followed by UV disinfection

HR: hydraulic retention time, SR: solids retention time.

resistant determinants in the environment as these elements efficient ERY-RGs among bacteria. (Baquero et al., 2008; Gao et al., 2015). Different ERY resistance genes are positioned in various genetic elements. Integrin holds ereA2, whereas macrolide phosphotransferases (mph) A, ereA, ereB and ermC are all positioned on plasmid and ermA, ermB, ermF are located on transposon (Pazda et al., 2019). The data in Table 3 provides detailed information about particular macrolide resistance genes detected in WWTP systems.

Gao et al. (2015) highlighted different concentration ranges of various ERY-RGs in wastewater. Significant differences were observed in their relative concentration. With a richness of 1.41×10^6 gene copies (GCs)/L in final effluent plus 2.63×10^8 GC/L in raw influent, the most abundantly detected genes were ere(A). The difference in quality of treatment methodology highly influences the presence

of ERY-RGs in different treatment plants. Michael et al. (2013) highlighted the resistant nature of ERY to biodegradation during biological treatment. Conventional activated sludge system seems very ineffective for removing ERY-RGs (Da Silva et al., 2006). Yuan et al. (2015) experimented chlorination for ERY-RG control reporting inefficiency of chlorination in antimicrobial control. ERY-RB displayed forbearance to chlorine dose up to 60 mg Cl₂ min/L, making chlorination unable to remove 40% of ERY-RG. As compared to chlorination UV disinfection has shown much positive results in reducing ERY-RGs (Guo et al., 2013).

ERY-RGs that include msr(A)/msr(B), erm(A) and erm(B) come under the category of movable genetic elements of human and poultry isolates. This implies that these could pass into clinically related bacteria. The continuous exposure to ERY can result in replacement of predominant

Table 3
ERY-RGs detected in world waters.

Country	Water body	Sample source	Prevalent antibiotic Resistant genes (erythromycin-resistance-methylase (erm))	References
China	Xiangjiang River	River water	ermB and ermC	(Xu et al., 2016)
	Pearl River and Estuary	River water/RS	ermB	(Chen et al., 2015)
Pakistan	Indus River	River water, canal, water, dam water, sewage water	ermB	(Khan et al., 2013)
Spain	Tordera River Basin	Biofilm	ermB	(Proia et al., 2016)
	Ter River	River water, hospital waste water, waste water treatment plant	ermB	(Rodríguez-mozaz et al., 2014)
Italy	Toce River	River water	ermB,	(Di et al., 2017)
USA	Cache La Poudre River Watershed	River water, river sediment;	ermF	(Garner et al., 2016)
Cuba	Almendares River	River water, river sediment;	ermE, ermB,	(Knapp et al., 2012)
	Almendares Rivera	River water, river sediment;	ermB, ermE,	(Werner and Bowen, 2011)

factions of microbial cloy composition to RGs detecting the mentioned ERY-RGs (Pala-Ozkok et al., 2014). Therefore the continuous ERY exposure poses a potentially serious health hazard (Gao et al., 2015).

7. Eco-toxicity and exposure to living biota

7.1. ERY concentration in living biota

Globally in the aquatic organisms' bioaccumulation of antibiotics has been observed. Historical approaches for defining bioaccumulation hazards and risks are dictated due to physicochemical properties of antibiotics and other pollutants (Chanterrie, 2004; Gaw et al., 2014). Owing to its nutritional benefits, seafood is the chief food but for local residents it is an important source of exposure to pharmaceuticals. The main seafood includes crabs, fish, mollusks and shrimps. A study reported the concentration of ERY-H₂O in mollusks to be 15,090 ng/g (wet weight), which was the highest out of total 17 antibiotics being identified. Whereas, in adult shrimps the ERY-H₂O concentration was reported to be varying from 2498 to 15,090 ng/g (Chen et al., 2014). Fish were more commonly studied as compared to other organisms, in them the ERY concentration was seen up to 2390 mg/kg compared to all other aquatic species which was from ND to 15,090 mg/kg. In municipal estuary environment in USA the fish were found to have much low maximal concentration i.e. 6.2 mg/kg (Schafhauser et al., 2018). In freshwater streams in the Pearl River Delta of China ERY concentration from fish liver sample was testified to be 2390 mg/kg (Zhao et al., 2016). In gill and liver of *Labeo rohita* fish elevated concentrations were recorded (Renuka et al., 2018).

7.2. Potential toxic effects

As per recent studies ERY pollution is posing an immense pressure on non-target aquatic organisms. It has affected their survival, growth rate and reproduction in a negative way (Isidori et al., 2005) especially, in cyanobacterium (Gonzalo et al., 2013), in algae (Eguchi et al., 2004; Wan et al., 2015), in different fish (Kim et al., 2009; Rocco et al., 2011; Rodrigues et al., 2019, 2016). In various animals such as rats due to

exposure of ERY, hepatotoxicity was observed (Pari and Murugan, 2004). In the quest to attain total ecological risk modelled by ERY exposure studies focused on computing hazard quotients (HQ), ecological impact is insignificant if HQ is less than unity (K. Ji et al., 2012) but it was calculated to be >1 i.e. 3.10. The effects of ERY on many species have been testified, both chronic and acute. As per a test of ERY introduction conducted on every life stage on medaka clearly shows the adverse effects on its survival and growth, depicting that 1000 mg/L contact to ERY is harmful for survival of medaka's larvae, juvenile and even the adult fish and the body measurements (length and weight) of a juvenile fish were erring from normal, hence proving ERY delays its growth as well (K. Ji et al., 2012). At a very minute concentration ERY tends to affect the water living photosynthetic organisms prominently even in mg/L concentrations (Gonzalo et al., 2013). *Selenastrum capricornutum*'s chlorophyll fluorescence, photosynthetic rate etc. at 60 mg/L has been affected (Liu et al., 2011) and at 1–10 mg/L *Synechocystis* sp. And *Lemna minor* were seen to have negative growth effects (Pomati et al., 2004). ERY in mussels had even provoked immunotoxic effects (Gust et al., 2012). Physiological and metabolic alterations can be excellent indicators of organism-contaminant interactions, and are the effects of environmental pollutants (Rodrigues et al., 2016). In the liver cells of *Oncorhynchus mykiss* report oxidative damage due to veterinary antibiotics (Rodrigues et al., 2019). In fish oxidative damage of gills, liver and kidney etc. is the result of the hyper reactive oxygen species (ROS) formulation, that can occur due to imbalance between the antioxidants in body and ROS production. Catalase (CAT), superoxide dismutase (SOD), lipid peroxidation (LPO) and glutathione peroxidase (GPx) are some of the significant antioxidant enzymes which via free radical scavenging property are meant to avert the inception of oxidative stress (Javed et al., 2017). They are reported to be the major toxicity evaluation biomarkers; detail is shown in Fig. 4.

ERY, a protein synthesis inhibitor belongs to macrolide antibiotics group. In prokaryotes gene expression, this antibiotic effectively inhibits the post-translation modification course. Gene expression of prokaryotes is in harmony with that of chloroplasts (Kasai et al., 2004). ERY not only is efficient in gene translation inhibition in chloroplast, but in algae also halts the membrane protein production in thylakoid. This

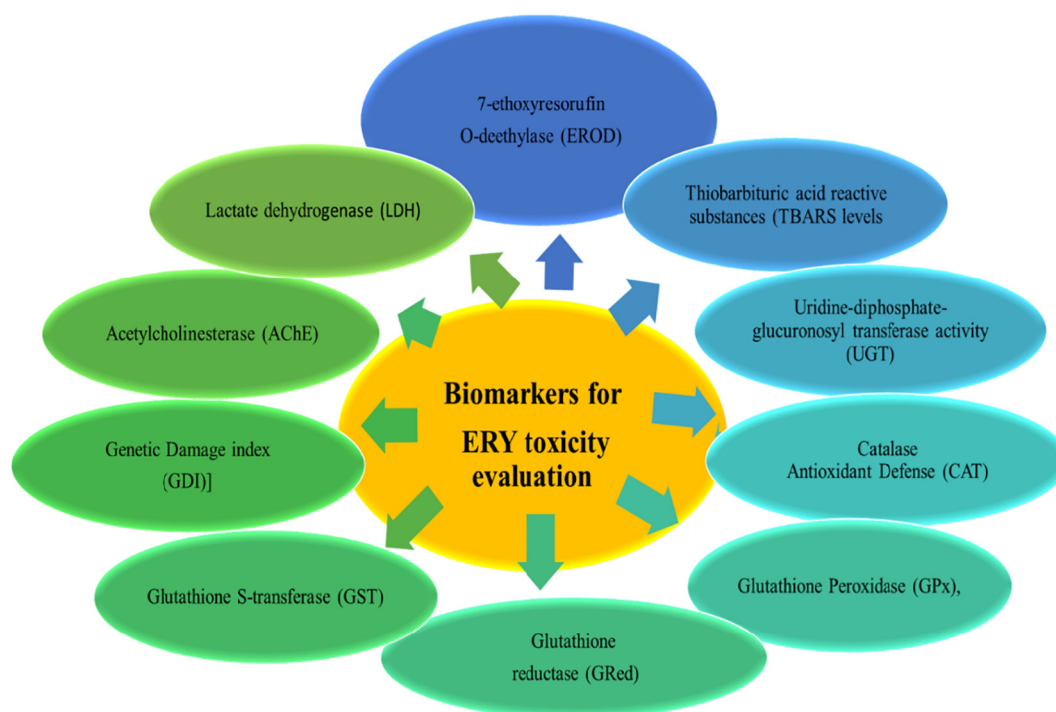


Fig. 4. ERY evaluation biomarkers in living biota.

can upshot in extreme bottom-up consequences on the higher trophic level e.g. herbivorous zooplanktons due to the concentration alteration of cyanobacteria and photosynthetic algae as they are the bases of the food webs. Hence, a good indicator for water quality as well as anthropogenic contamination is thought to be microalgae. According to most studies cyanobacteria “blue green algae” are vulnerable to the toxic effects of most of the pharmaceuticals being prokaryotes, possess higher sensitivity to antibiotics. Due to their sensitivity, cyanobacteria are to be used for effective testing of antimicrobials as explicitly indorsed by European Medicines Evaluation Agency (EMA). *Pseudokirchneriella subcapitata* (green alga) and cyanobacterium *Anabaena* CPB4337 had highly toxic effect of ERY (Gonzalo et al., 2013). Deadly effects of antibiotics to green algae could be associated to interference and reticence of

the chloroplast metabolism including interconnected protein formulation as well as photosynthesis procedures that interrupt the job of photosynthetic apparatus. In conclusion effecting cell growth, as eukaryotic green algae effecting chloroplast that belongs to the semi-autonomous organelle (Liu et al., 2011; Wan et al., 2015). ERY exposure to *Selenastrum capricornutum* affected its photosynthesis critically (Liu et al., 2011). ERY concentration is inversely proportional to photosynthesis rate, thus a decrease in fluorescence of chlorophyll was documented when it was exposed to 0.3 mg/L of ERY greatest. *Pseudokirchneriella subcapitata* and *Chlorella vulgaris* were 0.0366 mg/L and 33.9 mg/L when exposed to the EC₅₀ (72 h) of ERY, respectively (Eguchi et al., 2004) further detail given in Table 8. Thus, its emphases on the inhibitory effect of ERY on,

Table 4
Potential threats to living biota due to ERY pollution.

Test organisms	Specie group	Common name	Exposed organs	Testing biomarkers	Exposure dosage	Exposure duration	Major effects	References
<i>Sparus aurata</i> L	Marine fish	Gilt-head (sea) bream	Liver Gills	Detoxification Oxidative stress Lipid peroxidation Genotoxicity Neurotransmission Energy metabolism	AE/(0.3e323 mg/L.) CE/(0.7e8.8 mg/L.)	96 h and 28 days	EROD ⁰⁰ , GST ⁰⁰ , UGT ⁰⁰ AE/GPx(liver) ⁻⁻⁻ CE/GPx(gill) ⁺⁺ CE/TBARS ⁻⁻⁻ GDI ⁺⁺ AChE ⁰⁰ LDH ⁰⁰ STA/SOD ⁻⁻⁻ (gill) LTA/SOD ⁺⁺ (gill) SOD ⁺⁺ (liver) CAT ⁺⁺ (gill) CAT ⁺⁺ (liver) GPx ⁺⁺ (gill) GPx ⁺⁺ (liver) LPO ⁺⁺ (gill) LPO ⁺⁺ (liver)	(Rodrigues et al., 2018)
<i>Labeo rohita</i>	Fresh water fish	Roho labeo	Liver Gills	Superoxide dismutase Oxidative stress Guaiacol glutathione peroxidase Catalase Lipid peroxidation	10, 50, and 100 µg/L/	96 h and 35 days	STA/SOD ⁻⁻⁻ (gill) LTA/SOD ⁺⁺ (gill) SOD ⁺⁺ (liver) CAT ⁺⁺ (gill) CAT ⁺⁺ (liver) GPx ⁺⁺ (gill) GPx ⁺⁺ (liver) LPO ⁺⁺ (gill) LPO ⁺⁺ (liver)	(Renuka et al., 2018)
<i>Aedes aegypti</i> L.	Mosquito	Yellow fever mosquito	Larvae Eggs	Egg eclosion Larval mortality Pupation Emergency Fecundity	0, 0.02, 12, 24, 48, 96, 192 and 384 mg/L	–	Hyperpigmentation Blisters Survival decrease Progression delay	(Calma et al., 2018)
<i>Pseudokirchneriella subcapitata</i>	Green microalga	Raphidocelis subcapitata	–	Lipid peroxidation Total antioxidative capacity Ascorbate glutathione cycle Xanthophyll cycle The enzyme activities of catalase Superoxide dismutase Guaiacol glutathione peroxidase Glutathione-S-transferase	0 (control), 0.06, 0.12, 0.18, 0.24, 0.30 mg L ⁻¹	–	AsA ⁻⁻⁻ and GSH ⁻⁻⁻ , AGC ⁻⁻⁻ , XC ⁻⁻⁻ , CAT ⁻⁻⁻	(Nie et al., 2013)
<i>Anabaena</i> CPB4337	Cyanobacterium	Anabaena	–	–	EC50 = 0.35 _ 0.03 mg/L	–	–	(Gonzalo et al., 2013)
<i>Elliptio complanata</i>	Wild freshwater mussels	Naiads	Hemolymph	Immunotoxicity	10, 50, 250 and 1250 ng/L	–	LA– IC ++ IE ++ COX– ROS ++ Thiols – – PG – – E.Q.Y. PSII – –	(Gust et al., 2012)
<i>Chlamydomonas Reinhardtii</i>	Freshwater microalgae	–	–	oxidative damage photosynthetic apparatus	0.1, 1, 5, 10, 100 mg L _–	–	PG – – E.Q.Y. PSII – –	(Sendra et al., 2018)
<i>Phaeodactylum tricornutum</i>	Marine diatom	–	–	oxidative damage photosynthetic apparatus	0.1, 1, 5, 10, 100 mg L _–	–	PG – – E.Q.Y. PSII – –	(Sendra et al., 2018)
<i>Carassius auratus</i>	Fish	Crucian carp	Liver and brain	acetylcholinesterase 7-ethoxyresorufin O-deethylase superoxide dismutase	4, 20, and 100 µg/L	28 days	Biochemical disturbance of biological systems	(Liu et al., 2014)

Significant decrease—, significant increase++, no effect 00, CE = chronic exposure, AE = acute exposure EQY = effective quantum yield.

photophosphorylation, carbon assimilation, primary photochemistry, electron transport and many other photosynthesis-related processes and even with low exposure concentration of ERY can induce toxicity. ERY can not only affect algae in the environment by posing a high potential risk but also alter aquatic ecosystems, ecological balance, as indicated by these physiological responses. Other major toxic effects with their testing biomarkers are given in Table 4. For further advancement in understanding risks of pharmaceuticals including ERY on environment as well as on human health, future studies are looked-for, as the treatment groundwork and management systems are being executed slower than admittance of antibiotics, predominantly in developed regions.

8. Advance oxidation treatment

8.1. Ozonation

Antibiotic degradation through ozonation is a well-practiced technique due to its higher oxidation potential (2.07 V) (Iakovides et al., 2019; Pereira, 2007; Wang and Zhuan, 2020). Pollutant by means of this technique can either be degraded fully or can be converted to less toxic components (Blaney, 2014; Hembach et al., 2019). Through modulation of active functional group bacterial characteristics of an antibiotic can be passivated by ozone (O_3) e.g. N-ethoxime and dimethylamino groups of macrolides (Kurt et al., 2017). Ozonation can be an effective replacement on place of conventional biological treatment in a water treatment plant as the municipal wastes contain pollutants that require advance oxidation processes for their deletion therefore it can be used as tertiary treatment (Ikehata et al., 2007) as shown in Fig. 5. In the process of ozonation, time period and absorbance rate is observed to have an important position. In a study, decrease in all the solutions as well as parent antibiotic solution was observed with exception to ERY solution, as it absorbed a very little UV because of missing unsaturated C—C bond as compared to other antibiotics. A solution of five antibiotics when experimented under ozonation 99% degradation was reported in 10 min except ERY. ERY showed only 58% degradation in the initial time period whereas with the increase of time period to 45 min 99% degradation was recorded. The reason notified behind the slow ERY degradation is its complete saturated structure (Lin et al., 2009).

8.1.1. H_2O_2 mediated ozonation

Peroxonation principle is based on oxidizing radical generation through O_3 and H_2O_2 coupling (Lofrano et al., 2017). As notified by Ikehata et al. (2007) peroxonation could be double productive as compared to that of O_3 alone. O_3 water decomposition percentage is increased by the addition of H_2O_2 generating highly reactive $OH^\bullet$ radicals. This coupling enhances the economic feasibility of the entire process (Kurt et al., 2017). Several factors including minimum O_3 solubility, O_3 sensitivity as well as energy consumption play their part in limiting the process efficiency. In case of ERY degradation an opposite scenario is reported in a study where elevated ERY degradation efficiency was recorded using O_3 alone as compared to that of this combination. More than 94% of the removal efficiency of 13 antibiotics was observed by O_3 alone. Thus, considering O_3 as the main contributor for degradation of ERY during the peroxonation process (Kim and Tanaka, 2010).

8.1.2. Factors effecting ozonation

The O_3 oxidation mechanism is a multifarious practice proceeding in two folds: indirect oxidation via radical ($OH^\bullet$) formulation or direct reaction with dissolved O_3 . Generally, complete antibiotic mineralization is not possible via ozonation. Certain factors are involved in the process that can hinder the degradation mechanism e.g. CO_3^{2-} and HCO_3^- are $OH^\bullet$ radical scavengers and can constrain the subtraction of antibiotics formed throughout antibiotic decomposition process. Lofrano et al. (2017) showed the dose of ozone, or the pH of the medium, or the nature of the contaminant were the factors on which the allowance of both mechanisms right through the removal of a compound.

8.1.2.1. O_3 concentration. For antibiotic degradation, O_3 intensity has an important impact. With increase of O_3 concentration, the volumetric mass transfer coefficient along with mass transfer rate of O_3 escalates. O_3 decomposition rate is affected by Initial O_3 dose (Blaney, 2014), an important influence of O_3 dose on the removal rate of ERY is the result observed from the O_3 dose experiment. To effectively clarify the effect of O_3 dose on ERY oxidation by ozonation, an overall third-order reaction model was proposed. For 90% degradation of ERY using O_3 process a contact time of >10 min and O_3 dose of at least 6 mg/L was necessary as reported in a study (Kim and Tanaka, 2010). With an O_3 dose of >2 mg/L it has been reported that the amount of ERY oxidized was 90–99% (Ling, 2011).

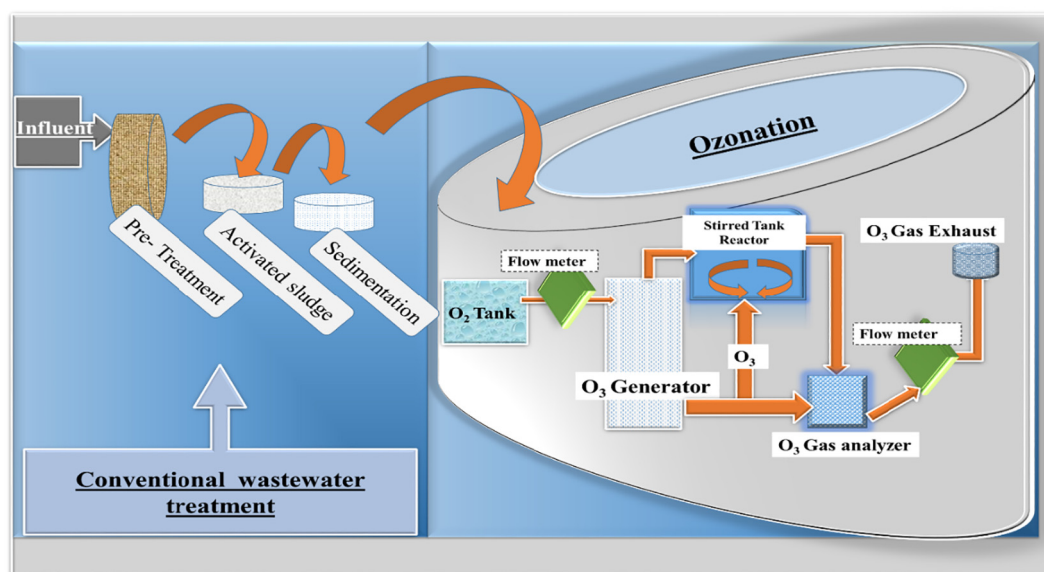


Fig. 5. Ozonation process fixed as tertiary treatment (Hembach et al., 2019).

8.1.2.2. pH. The decrease of pH during ozonation imposes a negative impact of OH• generation (Blaney, 2014). A study reported that direct ozonation prevails under acidic conditions (pH < 4), normally. But in case of ERY degradation at pH 3 the degradation was 19% and at pH 5 a slight high degradation is observed (ERY removal = 25%). With increasing solution pH to 8 kapp was elevated, ERY degradation was affected kinetically (Cuerda-correa et al., 2020). Oxidation of the organic pollutants occurs both by hydroxyl radical and O₃ molecule in alkaline solution. Similar and fast removal rates are observed at pH 7 and 11 as compared to pH 3. At high pH as the concentration of O₃ decreases, its involvement in ERY degradation is also abridged. From elevated hydrolysis of O₃, concentration of OH• radical is increased thus compensating the degradation rate. ERY lacks unsaturated bonds while other compounds contain minimum 2 C—C bonds that are unsaturated. Reduced O₃ at high pH does not hinder the removal of ERY but it assists O₃ and OH• radical production in other unsaturated compounds. At the medium pH of 7 the degradation rate of ERY is closest to the others also suggests that at alkaline or neutral ranges of pH the degradation of ERY is enviable (Lin et al., 2009).

8.1.3. Sulfate radical based AOPs

Due to higher oxidation potential, greater pollutant oxidizing efficiency and selectivity with wide pH range sulfate radical based AOPs have received increasing attention for antibiotic degradation. SO₄•[−] are formulated by the activation of persulfate (PS) or peroxymonosulfate (PMS) (Guerra-Rodríguez et al., 2018) as details are given in Table 5. Sulfate produced radicals react with organic pollutants causing either fractional or all-inclusive mineralization (Wang and Wang, 2018). Lee et al. (2015) clearly shows the mineralization, either partial or complete as an outcome of the SO₄•[−] as well as organic chemical reactions. The three ways by which SO₄•[−] reacts are: hydrogen abstraction, direct electron transfer and addition to aliphatic double bonds (Matzek and Carter, 2016). For contaminant degradation the mechanism used varies pointedly as it depends upon nature of contaminant and free radical and the organic compounds of different kinds can be degraded by OH• and SO₄•[−]. According to reports a fair difference was seen in the reaction mechanism of and OH• radical. OH• may involve hydrogen-atom abstraction reactions along with electron-transfer reactions whereas, only electron transfer reactions are involved in SO₄•[−] (Kurt et al., 2017). Before being activated for contaminant degradation the PS or PMS anion may be transported further in the subsurface as it is more stable (Wang and Wang, 2018).

8.1.3.1. Radical formation. Two SO₄•[−] are formed along with cleavage of the peroxide bond by the energy given to PS anion via UV light, ultrasound and heat. As depicted from the Table 6, consecutively with an electron donor from the radiolysis of water or a transition metal, PS can undergo redox reaction. From SO₄•[−] reactions certain secondary radicals propagate which boost the effect of activated PS degradation reactions. OH• are the primary reactive species under basic conditions and

are formed at all pH when SO₄•[−] react with water (Wang and Wang, 2018). At neutral pH SO₄•[−] and OH• contribute equally in reactions whereas the dominant reactive species at pH < 7 are SO₄•[−] as shown in Table 6. Due to the fact that PS or PMS anion has lower oxidation potential so without activation when there will be a reaction between the PS or PMS anion and some organic pollutant, the degree of removal will be a smaller than that obtained using activated one (Guerra-Rodríguez et al., 2018). SO₄•[−] formed as a product of reaction amid ERY and PS, may create ERY radicals (as shown in the Fig. 6) or promulgate a secondary reaction with time decomposing the envisioned antibiotic. The complete degradation rates depend on a composite combination of SO₄•[−] chain propagation and terminating reaction (Wang and Wang, 2018; Zhang et al., 2019b). Many studies were reported using activated PS with various sources for ERY degradation. Detail is given Table 7.

8.1.4. Activation sources

8.1.4.1. Heat activation. Several reports witness thermo activation of PS and PMS (Huang et al., 2005; Ji et al., 2016). Fission of O—O bond is recorded due to energy input that ultimately results in radical production. Radical generation is directly proportional to the temperature provided. Greater the temperature greater will be the radical generation therefore, enhancing the pollutant degradation (Guerra-Rodríguez et al., 2018). 99% degradation of ERY in from ERY mycelial dreg experiments using heat activated PS oxidation technique was recorded in alkaline environment keeping temperature up to 80 °C under controlled conditions (Zhang et al., 2019b). For extracting ERY containing organic matter, mycelial cells, and large amount of ERY (0.64 g/kg) a type of antibiotic mycelial residue (AMR) was used known as ERY mycelial dreg (Cai et al., 2020). ERY mycelial dreg produce due to its high moisture content is used as a source of nutrient, making it complicated to handle the operations under heat-activated PS oxidation conditioning, Zhang et al. (2019b) studied the ERY mycelial dreg dewaterability and reported that the formation of crystal structures, improved ERY mycelial dreg filtration dewaterability, destruction of ERY mycelial dreg flocs, EPS, mycelial cells are result of oxidation by SO₄•[−] and HO•. For destroying ARGs and for restricting the spread of antibiotic resistance mycelial dreg effective treatments are highly looked-for. Zhang et al. (2020) experimented heat-activated PS oxidation effects on the treated ERY fermentation residue (EFR) on chemical properties of soil as well as ERY dissipation and ERY-RG evolution. No resistance gene enrichment in the soil was recorded by heat-activated PS oxidation treated EFR. Therefore, this treatment technique was observed to be very efficient to treat ERY resistance risk. To evade any kind of economic burden or leakage this remediation technology is preferably applied in-situ (Devi et al., 2016).

8.1.4.2. Radiation activation. PMS and PS can be activated by radiations (Guerra-Rodríguez et al., 2018) such as gamma ray (Chu et al., 2019), and ultraviolet (UV) (Zhou et al., 2020) in the water and wastewater, for the degradation of ERY. The most cost-effective and benign way to activate PS is considered to be ultraviolet radiation. The energy absorbance with wavelength 254 nm is considered to be most used wavelength (Wang and Wang, 2018). When chemical oxidants such as PS, PMS or H₂O₂ are combined into UV system its degradation efficiency increases pointedly. PS and H₂O₂ are activated by UV radiations producing SO₄•[−] and HO• radicals and for the degradation of ERY in wastewater systems these radicals show a high potential. To incapacitate ERY-resistant bacteria and degrade ERY in sewage the photochemical process using UV activated by S₂O₈^{2−} was studied by (Michael-Kordatou et al., 2015). Results indicated that within 90 min 100% of ERY removal could be achieved with preliminary ERY dilution of 0.1 mg/L by UV activated PS system; whereas with only UV irradiations the yield was only 34%. Chu et al. (2019) experimented ERY degradation by means of gamma rays. PS-gamma combination resulted in improved ERY degradation. With 85% utilization of PS the ERY degradation percentage was

Table 5
PS and PMS profile.

Properties	PS	PMS
Formula	S ₂ O ₈ ^{2−}	HSO ₅ [−]
Shape	Colorless or white crystal	White solid powder
Water solubility	730 g L ^{−1}	>250 g L ^{−1}
Water solution	Acidic	Acidic
Structure	Symmetrical	Asymmetrical
o-o bond distance	1.497 Å	1.453 Å
Bond energy	140 KJ/Mol	140–213.3 KJ/Mol
Redox potential	2.01 V	1.82 V
Common	Sodium PS	Oxzone
experimentation use	(Na ₂ S ₂ O ₈) potassium PS (K ₂ S ₂ O ₈)	(KHSO ₅ · 0.5KHSO ₄ · 0.5K ₂ SO ₄)

Table 6

Complete mechanism of ozonation, sulfate based AOP and photocatalysis.

AOPs	Mechanism
Catalytic ozonation Ozonation (O ₃)	$\begin{aligned} \text{O}_3 + \text{H}_2\text{O} &\rightarrow 2\text{OH}^\bullet + \text{O}_2 \\ \text{O}_3 + \text{OH}^- &\rightarrow \text{O}_2^{\bullet-} + \text{HO}_2^\bullet \\ \text{O}_3 + \text{HO}_2^\bullet &\rightarrow 2\text{O}_2 + \text{OH}^\bullet \\ \text{O}_3 + \text{OH}^\bullet &\rightarrow \text{O}_2 + \text{HO}_2^\bullet \\ 2\text{HO}_2^\bullet &\rightarrow \text{O}_2 + \text{H}_2\text{O}_2 \\ \text{H}_2\text{O}_2 &\rightarrow \text{HO}_2^- + \text{H}^+ \\ \text{HO}_2^- + \text{O}_3 &\rightarrow \text{HO}_2^\bullet + \text{O}_3^- \\ \text{HO}_2^\bullet &\rightarrow \text{H}^+ + \text{O}_2^- \\ \text{O}_2^{\bullet-} + \text{O}_3 &\rightarrow \text{O}_2 + \text{O}_3^- \\ \text{O}_3^- + \text{H}^+ &\rightarrow \text{HO}_3 \\ \text{HO}_3^\bullet &\rightarrow \text{HO}^\bullet + \text{O}_2 \\ \text{HO}^\bullet + \text{O}_3 &\rightarrow \text{HO}_2^\bullet + \text{O}_2 \\ 2\text{H}_2\text{O}_2 + \text{O}_3 &\rightarrow 2\text{HO}^\bullet + \text{O}_3^- \\ \text{O}_3 + \text{H}_2\text{O} &\rightarrow [2\text{OH}^\bullet] + \text{O}_2 \rightarrow \text{H}_2\text{O}_2 + \text{O}_2 \\ 2\text{H}_2\text{O}_2 + \text{O} &\rightarrow 2\text{HO}^\bullet + \text{O}_3^- \end{aligned}$
O ₃ /H ₂ O ₂	
O ₃ /H ₂ O ₂ /UV	
Activated persulfate oxidation Persulfate reaction mechanism Formation of sulfate, hydroxyl, and other radical/water radiolysis	$\begin{aligned} \text{S}_2\text{O}_8^{2-} + \text{energy input} &\rightarrow 2\text{SO}_4^{\bullet-} \\ \text{S}_2\text{O}_8^{2-} + \text{e}^- &\rightarrow \text{SO}_4^{\bullet-} + \text{SO}_4^- \\ \text{SO}_4^- + \text{H}_2\text{O} &\rightarrow \text{SO}_4^{\bullet-} + \text{OH}^\bullet + \text{H}^+ \\ \text{SO}_4^- + \text{OH}^- &\rightarrow \text{SO}_4^{\bullet-} + \text{OH}^\bullet \\ \text{SO}_4^- + \text{OH}^\bullet &\rightarrow \text{HSO}_5^- \\ 2\text{OH}^\bullet &\rightarrow \text{H}_2\text{O}_2 \\ \text{S}_2\text{O}_8^{2-} + \text{R} &\rightarrow \text{SO}_4^- + \text{SO}_4^{\bullet-} + \text{R} \\ \text{S}_2\text{O}_8^{2-} + \text{R} &\rightarrow 2\text{SO}_4^- + \text{R}^\bullet \end{aligned}$
Sulfate degradation mechanisms	
Degradation reactions by activation type without Iron-activated persulfate for analyte degradation persulfate	$\begin{aligned} \text{S}_2\text{O}_8^{2-} + \text{Fe}^{2+} &\rightarrow \text{Fe}^{3+} + \text{SO}_4^{\bullet-} + \text{SO}_4^- \\ \text{Fe}^0 + \text{S}_2\text{O}_8^{2-} &\rightarrow \text{Fe}^{2+} + 2\text{SO}_4^{\bullet-} \\ \text{SO}_4^{\bullet-} + \text{Fe}^{2+} &\rightarrow \text{Fe}^{3+} + \text{SO}_4^- \\ \text{S}_2\text{O}_8^{2-} + \text{heat} &\rightarrow 2\text{SO}_4^- \\ \text{S}_2\text{O}_8^{2-} + \text{h}\nu &\rightarrow 2\text{SO}_4^- \end{aligned}$
Heat-activated persulfate for analyte degradation persulfate UV light activated persulfate for analyte degradation persulfate Persulfate activation methods Electrochemically-activated persulfate for analyte degradation	$\begin{aligned} \text{Fe}^0 &\rightarrow \text{Fe}^{2+} + \text{e}^- \\ \text{Fe}^{3+} + \text{e}^- &\rightarrow \text{Fe}^{2+} \\ \text{S}_2\text{O}_8^{2-} + 2\text{H}_2\text{O} &\rightarrow 2\text{SO}_4^- + \text{SO}_4^{\bullet-} + \text{O}_2^{\bullet-} + 4\text{H}^+ \\ \text{AC surface} - \text{OOH} + \text{S}_2\text{O}_8^{2-} &\rightarrow \text{SO}_4^{\bullet-} + \text{AC surface} - \text{OO}^\bullet + \text{HSO}_4^- \\ \text{AC surface} - \text{OH} + \text{S}_2\text{O}_8^{2-} &\rightarrow \text{SO}_4^{\bullet-} + \text{AC surface} - \text{O}^\bullet + \text{HSO}_4^- \\ \text{S}_2\text{O}_8^{2-} \xrightarrow{\gamma\text{-rays/EB}} &2\text{SO}_4^{\bullet-} \\ \text{HSO}_5^- \xrightarrow{\gamma\text{-rays/EB}} &\text{SO}_4^{\bullet-} + \text{.OH} \end{aligned}$
Base-activated persulfate Ultra-sonication, activated-carbon, and other methods of persulfate activation for analyte degradation Activate persulfate by gamma rays	
Factors affecting effectiveness of oxidation process pH	$\begin{aligned} \text{S}_2\text{O}_8^{2-} + 2\text{H}_2\text{O} &\rightarrow \text{SO}_4^- + \text{O}_2^- + 4\text{H}^+ \\ \text{SO}_4^- + \text{HO}^- &\rightarrow \text{SO}_4^{\bullet-} + \text{HO}^\bullet \\ \text{Fe}^{3+} + \text{H}_2\text{O} &\rightarrow \text{Fe}(\text{OH})^{2+} + \text{H}^+ \\ \text{Fe}^{3+} + 2\text{H}_2\text{O} &\rightarrow \text{Fe}(\text{OH})_2^{2+} + 2\text{H}^+ \\ 2\text{Fe}^{3+} + 2\text{H}_2\text{O} &\rightarrow \text{Fe}_2(\text{OH})_2^{4+} + 2\text{H}^+ \\ \text{HO}^\bullet + \text{O}_2^- &\rightarrow \text{O}_2 + \text{HO}^- \\ \text{SO}_4^- + \text{O}_2^- &\rightarrow \text{SO}_4 + \text{O}_2 \\ \text{H}_2\text{O}_2 + \text{HO}^\bullet &\rightarrow \text{H}_2\text{O} + \text{HO}_2^\bullet \\ 2\text{H}_2\text{O}_2 &\rightarrow \text{O}_2 + 2\text{H}_2\text{O} \\ 2\text{HO}_2^\bullet &\rightarrow \text{H}_2\text{O}_2 + \text{O}_2 \\ \text{HO}_2^\bullet + \text{H}_2\text{O}_2 &\rightarrow \text{HO}^\bullet + \text{H}_2\text{O} + \text{O}_2 \\ \text{Fe}^{2+} + \text{H}_2\text{O}_2 &\rightarrow \text{Fe}^{3+} + \text{HO}^\bullet + \text{HO}^- \\ \text{Fe}^{3+} + \text{H}_2\text{O}_2 &\rightarrow \text{Fe}^{2+} + \text{HO}_2^\bullet + \text{H}^+ \\ \text{Fe}^{2+} + \text{HO}^\bullet &\rightarrow \text{Fe}^{3+} + \text{HO}^- \\ \text{Fe}^{3+} + \text{HO}_2^\bullet &\rightarrow \text{Fe}^{2+} + \text{O}_2 + \text{H}^+ \\ 2\text{HSO}_5^\bullet &\leftrightarrow \text{HO}_3\text{SOOOOSO}_3\text{H} \rightarrow \text{SO}_4^{\bullet-} + \text{O}_2 + 2\text{H}^+ \\ \text{Ag}^+ + \text{S}_2\text{O}_8^{2-} &\rightarrow \text{Ag}^{3+} + 2\text{SO}_4^{\bullet-} \\ \text{CO}^{2+} + \text{SO}_4^- &\rightarrow \text{CO}^{3+} + \text{SO}_4^{\bullet-} \\ \text{HSO}_3^- + \text{HO}^\bullet &\rightarrow \text{SO}_3^- + \text{H}_2\text{O} \\ \text{HSO}_3^- + \text{SO}_4^- &\rightarrow \text{SO}_3^- + \text{HSO}_4^- \\ \text{HO}^\bullet + \text{HO}^\bullet &\rightarrow \text{H}_2\text{O}_2 \\ \text{SO}_4^- + \text{SO}_4^- &\rightarrow \text{S}_2\text{O}_8^{2-} \\ \text{SO}_4^- + \text{S}_2\text{O}_8^{2-} &\rightarrow \text{S}_2\text{O}_8^- + \text{SO}_4^{\bullet-} \\ \text{SO}_4^- + \text{H}_2\text{O} &\rightarrow \text{SO}_4^{\bullet-} + \text{H}^+ + \text{HO}^\bullet \\ \text{SO}_4^- + \text{HO}^- &\rightarrow \text{SO}_4^{\bullet-} + \text{HO}^\bullet \\ \text{HCO}_3^- + \text{HO}^\bullet &\rightarrow \text{CO}_3^- + \text{H}_2\text{O} \\ \text{CO}_3^{2-} + \text{HO}^\bullet &\rightarrow \text{CO}_3^- + \text{HO}^- \\ \text{SO}_4^- + \text{HCO}_3^- &\rightarrow \text{CO}_3^- + \text{SO}_4^{\bullet-} + \text{H}^+ \\ \text{SO}_4^- + \text{Cl}^- &\rightarrow \text{Cl}^\bullet + \text{SO}_4^{\bullet-} \\ \text{HO}^\bullet + \text{Cl}^- &\rightarrow \text{Cl}^\bullet + \text{HO}^- \\ \text{Cl}^\bullet &\rightarrow \text{Cl}^- \rightarrow \text{Cl}_2^- \\ \text{SO}_4^- + \text{CO}_3^{2-} &\rightarrow \text{CO}_3^- + \text{SO}_4^{\bullet-} \\ \text{Cl} + \text{H}_2\text{O} &\rightarrow \text{HO}^\bullet + \text{HCl} \end{aligned}$
Oxygen concentration	
Reaction intermediates and other scavengers	
Carbonates, bicarbonates and chlorides	

(continued on next page)

Table 6 (continued)

AOPs	Mechanism
Photocatalysis	$\text{Cl}_2^- + \text{H}_2\text{O} \rightarrow \text{HO} \cdot + 2\text{Cl}^- + \text{H}^+$
Photocatalytic processes	$\text{Cl}_2^- + \text{HO}^- \rightarrow \text{HO} \cdot + 2\text{Cl}^-$
Photoexcitation	Photocatalytic reactions
Ionization of water	$\text{Photocatalyst} + h\nu \rightarrow e_{cb}^- + h_{vb}^+$
Ionization of water	$h^+ + \text{H}_2\text{O} \rightarrow \text{H}^+ + \text{OH} \cdot$
Hydroxyl radical production	$h^+ + \text{H}_2\text{O} \rightarrow \text{H}^+ + \text{OH} \cdot$
Production of super oxide radicle	$h^+ + \text{OH}^- \rightarrow \text{OH} \cdot$
Hydrogen peroxide production	$e^- + \text{O}_2 \rightarrow \text{O}_2^{\bullet -}$
Production of hydroxyl radical and ion	$\text{O}_2^{\bullet -} + \text{H}_2\text{O} + \text{H}^+ \rightarrow \text{H}_2\text{O}_2 + \text{OH}^-$
Degradation	$\text{H}_2\text{O}_2 + e^- \rightarrow \text{OH} \cdot + \text{OH}^-$
	$\text{OH} \cdot + \text{ERY} \rightarrow \text{Intermediates} + \text{H}_2\text{O} + \text{CO}_2$
	$\text{O}_2^{\bullet -} + \text{ERY} \rightarrow \text{Intermediates} + \text{H}_2\text{O} + \text{CO}_2$

recorded to be 15% but with the increase in gamma radiation to 6 kGy complete ERY degradation was obtained making this combination quite effective. In gamma-PS system an elevated yield of acetic acid and formic acid was observed. LC-MS reading proved the destruction of 14 membraned ERY ring. Demonstrated by Shen et al. (2019) gamma irradiation at 30 kGy revealed to ERY- thiocyanate fermentation wastes the removal efficiency of macrolide ARGs and ERY possibly reach

90–95% and 56%, respectively. Reaction time or high dose is needed for reaching the high ERY mineralization extent making it less economic. A positive point in using gamma irradiations is that it requires room temperature to operate. Both ARGs and ERY were efficiently eliminated through radical oxidation and reduction process formed in water radiolysis.

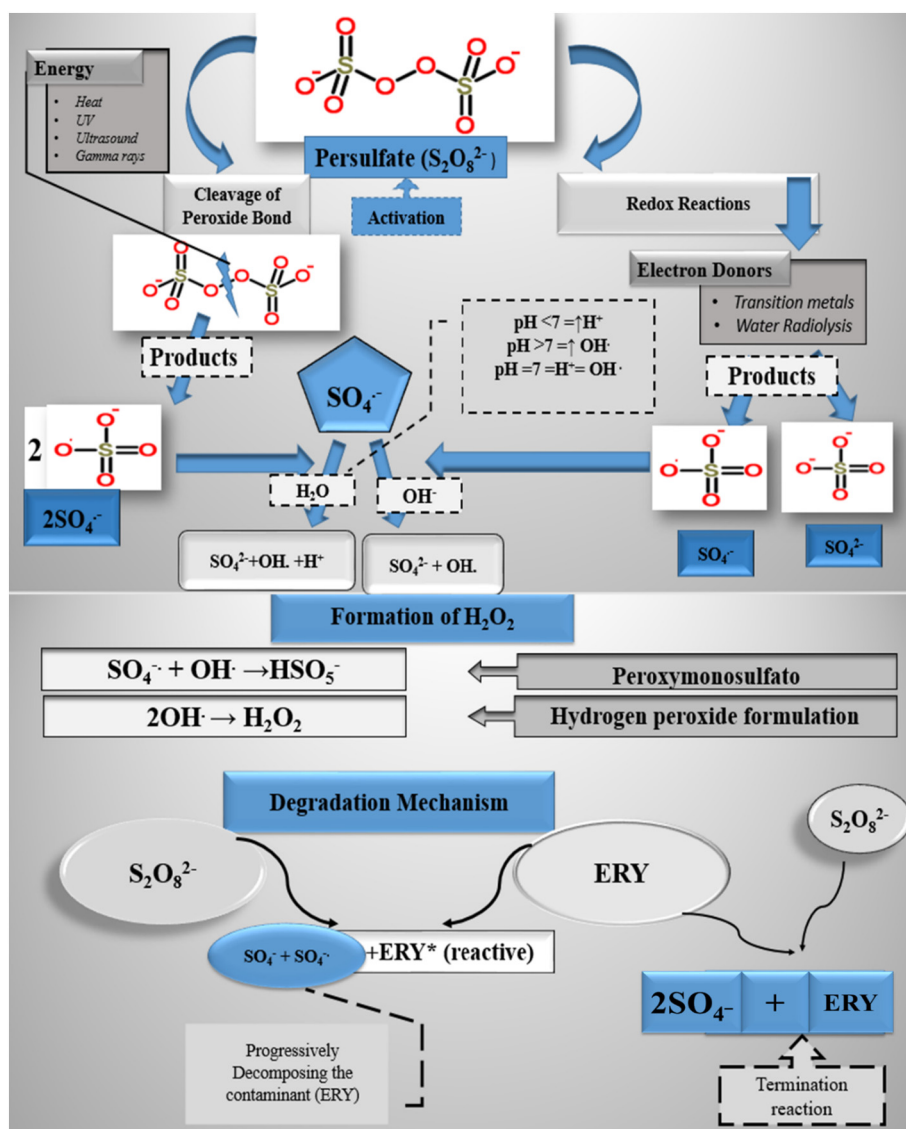


Fig. 6. Complete overview of ERY degradation by Sulfate radical based AOPs.

Table 7
ERY degradation by Sulfate based AOPs.

Contaminant	ERY concentration	Activation process	Activator dosage	Oxidant	Oxidant conc.	pH	Temperature	Retention time (min)	Wavelength	Removal efficiency	Ref.
EMD	0.64 ± 0.02 g/kg	Heat	-	PS	50 g/kg	5.93 ± 0.03	80 °C	120	-	99.0%	(Zhang et al., 2019a)
EFr	640 mg/kg	Heat	-	PS	25 g/kg	-	80 °C	120	-	97.39%	(Zhang et al., 2020)
ERYTcF	538 mg/kg	Ionizing radiation	30 kGy	Radioactivity + H ₂ O ₂	-	8.6–9.13	19–22 °C	-	-	38–56%	(Shen et al., 2019)
ERY	0.10 mM, 0.05 Mm, 0.02 mM	Gamma rays	0.8 kGy min ⁻¹	PS	1 mm	-	20–22 °C	-	-	79%	(Chu et al., 2019)
ERY	1 mg/L	Iron activated	22.4 mg/L	PS + H ₂ O ₂	0.1 mm	-	25 ± 2 °C	-	-	100%	(Li et al., 2017)
ERY	100 mg/L ¹	UV	-	Sodium PS	0–50 mg/L ¹	8	22 ± 2 °C	90	254 nm	37% and 63%	(Michael-Kordatou et al., 2015)
EMD	0.64 ± 0.02 g/kg	Heat	-	PS	50 g/kg	5.93 ± 0.03	80 °C	60	-	-	(Zhang et al., 2019a)

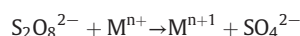
EMD = erythromycin mycelial dreg, EFR = erythromycin fermentation residue, ERYTcF = erythromycin thiocyanate fermentation.

the increase of 20 °C keeping the retention time 120 min for both. By means of heat-activation PS oxidation 99% ERY removal was reported at 80 °C.

8.1.5.2. Effect of PS dosage. As reported in study conducted by Zhang et al. (2019b), increasing PS dosage per kg ERY resulted in elevated degradation efficiency. 1 g PS/kg ERY mycelial dreg, 5 g PS /kg ERY mycelial dreg, 10 g PS /kg ERY mycelial dreg and 25 g PS /kg ERY mycelial dreg contained by retention time of 120 mins was 72.8%, 79.1%, 90.5% and 99.0% respectively under heat activation mechanism therefore proving that PS dosage increase enhances contaminant degradation efficiency.

8.1.5.3. Effect of pH. Alkaline conditions are reported to be the most favorable ones for the degradation of organic contaminants by means of SO₄•⁻ radicals. Transformation of PS and PMS to SO₄•⁻ radical is possible under alkali environment (Wang and Wang, 2018). A study reported the maximum decomposition of PMS at pH 9.0. The SO₄•⁻ radicals can be transferred to HO• radicals in extreme alkaline conditions by its reaction with HO ion (Devi et al., 2016). Chu et al. (2019) investigated that ERY degradation by means of SO₄•⁻ radicals is possible between 3 and 10.5 but alkaline pH elevated the degradation as compared to the acidic one. On other hand Guerra-Rodríguez et al. (2018) reported that neutral or acidic pH is more effective for SO₄•⁻ generation and decrease in its production is recorded in basic medium. Still for ERY degradation alkaline medium was observed to be more feasible as compared to that of acidic (Zhang et al., 2019b).

8.1.5.4. Radical scavengers. Radical specie formulation is confirmed by conducting various radical scavenger studies (Wang and Wang, 2018). Studies used EtOH and t-BuOH that react with radicals. Studies proved 50 times higher reaction constant of EtOH with OH• as compared to that with SO₄•⁻ (Michael-Kordatou et al., 2015). Without alcohols 99% ERY degradation was recorded due to OH• and SO₄•⁻ synergistic effect. This implies that, slower degradation was recorded due to their scavenging properties of EtOH and t-BuOH. An obvious ERY degrading **8.1.4.3. Metal activation.** Transition metals can be a very effective PS or PMS activation source (Guerra-Rodríguez et al., 2018). Metals such as cobalt, silver, manganese, copper, zinc and iron are used for 1 electron transfer which intends to activate PS or PMS (Ghanbari and Moradi, 2017; Rodr et al., 2019) (shown in Fig. 7) and as a result producing SO₄•⁻ as shown in equation:



where M being any transition metal. Iron due to its properties of being non-toxic, cost effective and environmental friendliness, is used extensively. Two SO₄•⁻ are produced with activation of PS whereas, activation of PMS produces one OH• and one SO₄•⁻. PMS has an advantage of reaction with transition metals producing sulfur pentoxide radical (SO₅•⁻). Well, it is less reactive than SO₄•⁻ but can effectively target the pollutant (Guerra-Rodríguez et al., 2018) as given in the Fig. 6. The SO₄•⁻ radical can be made more effective than other transition metals by its reaction with ferrous iron (Fe²⁺), mechanism is described in the table. Swift and massive decomposition of H₂O₂ is facilitated after we add persulfate for the effective oxidation of ERY which accompanies the formation of surface-associated Fe (II) on the ZVI surface (Li et al., 2017).

8.1.5. Factors effecting sulfate radical based AOPs

8.1.5.1. Effect of temperature. Contaminant degradation could increase with the increase in temperature by elevating the radical production facilitating PS decomposition. Zhang et al. (2019b) reported only 44.8% ERY degradation when the temperature was kept at 40 °C. ERY removal increased with the increase in temperature reaching up to 71.9% with

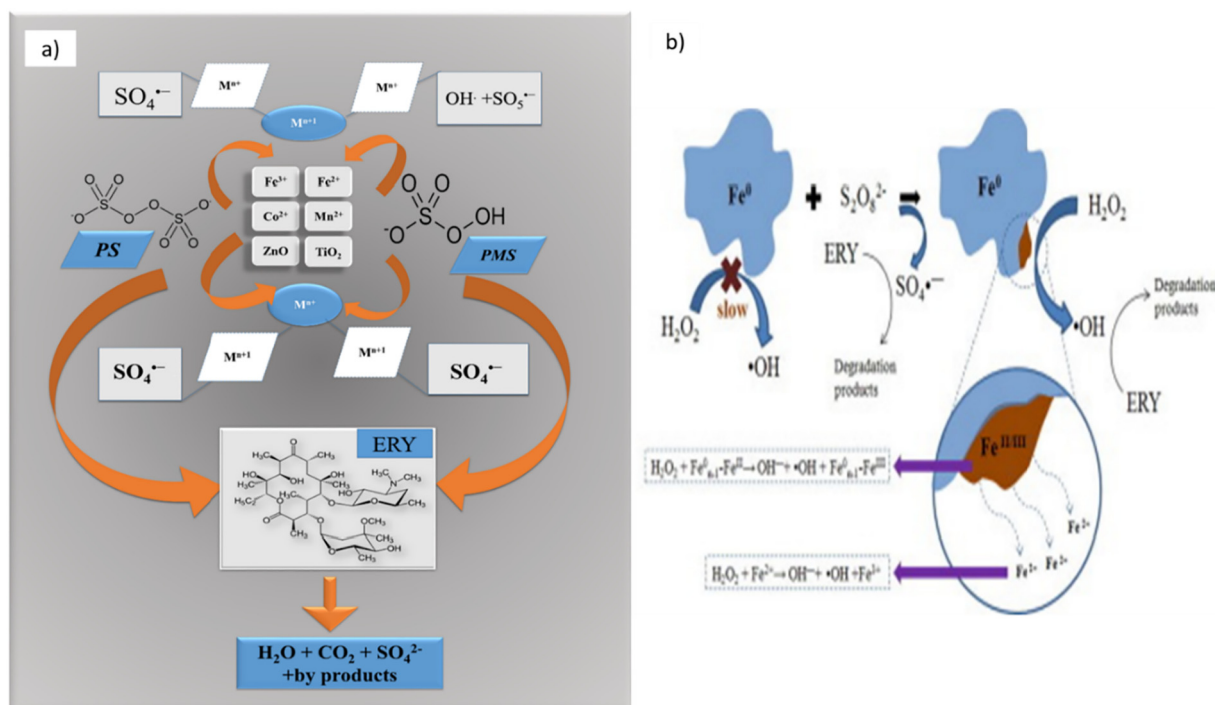


Fig. 7. a) The scheme of PS and PMS activation by transition metals b) oxidation of erythromycin by persulfate activated iron powder-H₂O₂ system (Li et al., 2017) copyrights 2017 reprinted with permission of Elsevier.

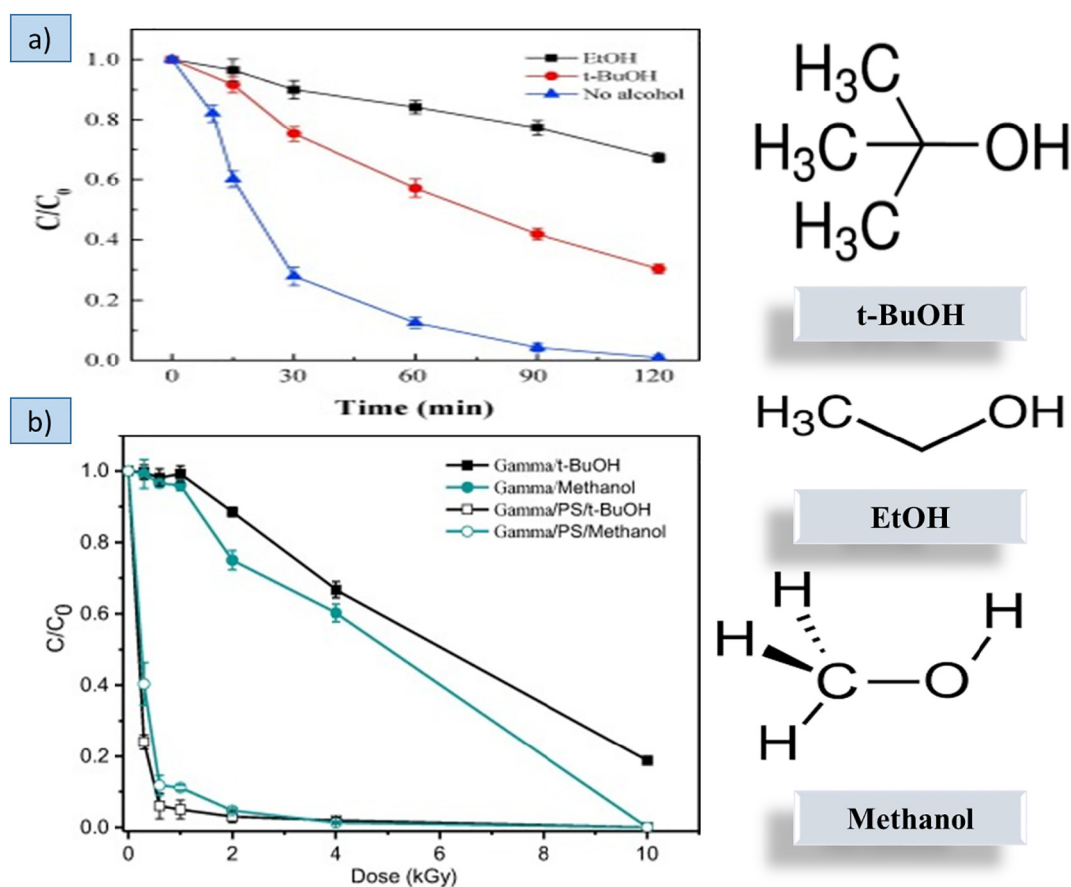


Fig. 8. a) The scavenging tests during the heat-activated PS oxidation (Zhang et al., 2019b) copyrights 2019, reprinted with permission of Elsevier b) Effects of different scavengers (500mMt-BuOH and methanol) on ERY degradation by gamma irradiation and Gamma/PS system (Chu et al., 2019) copyrights 2019, reprinted with permission of Elsevier.

inhibition was reported due to presence of EtOH and t-BuOH, as only 32.8% degradation was recorded in the presence of EtOH whereas, 69.6% ERY degradation in the presence of t-BuOH (Zhang et al., 2019b) as shown in Fig. 7. Chu et al. (2019) reported ERY degradation decrease by 1.4–1.7 times in Gamma/PS system due to t-BuOH and methanol as shown in Fig. 8.

8.2. Photocatalytic degradation of ERY

8.2.1. Photocatalytic mechanism

One of the most studied processes for elimination of pollutants from the aquatic ecosystem is photocatalysis (Chen et al., 2019; Dong et al., 2019; Liu et al., 2019; Tammam et al., 2017; Thangavel et al., 2016; Xu et al., 2018). Photocatalysis include the formation of oxidizing agents in the presence of a catalyst under solar irradiation (Gholami et al., 2020; Ohtani et al., 2013; Qu, 2019). The antibiotics are mineralized to simpler compounds like H_2O , CO_2 and simpler mineral acids (Alatrache et al., 2010; Gao et al., 2016; Gholami et al., 2020; Gurke et al., 2015; Kumar et al., 2019; Malakootian et al., 2019; Wu et al., 2020). The process involves the light source illuminating the semiconductor material. The band gap of the material used should be less than the energy provided. Electrons transfer towards the conduction band (cb) from the valence band (vb) forming the holes (h^+). The holes further react with the water (H_2O) or hydroxyl ions (OH^-) to produce the hydroxyl radical ($OH^\bullet$). Superoxide ions are generated in the cb as the electrons react with O_2 that ultimately form $OH^\bullet$ and hydrogen peroxide (H_2O_2) when placed in acidic medium. Step by step degradation equations are given in Table 6. It has been observed that irradiation can also decompose H_2O_2 to $OH^\bullet$. $OH^\bullet$, $O_2^{\bullet-}$ and h^+ react with ERY, promoting its consequent degradation (Malakootian et al., 2019; Nguyen et al., 2019) as shown in Fig. 9. Photocatalysts plays a vital role in these light

induced reactions (Schneider et al., 2014). The major qualities to be observed before choosing the ideal photocatalyst are its economic feasibility, nontoxicity, stability, availability and photoactivity. Studies show that instead of a single photocatalyst the composites can play an effective role. As observed in degradation of ERY the efficiency of Znpc-TiO₂ was double then that of single TiO₂. The ERY degradation percentage reached to almost 74.21% using Znpc-TiO₂ whereas on the other hand the degradation percentage was only 31.57% using TiO₂ (Vignesh et al., 2014a,b). In the case of using composite the first catalyst loaded on the surface of the other can create mobile electrons by its excitation that can be injected into the cb of the second catalyst. Those electrons can react with the surface adsorbed oxygen and this process can enhance the production of reactive oxygen species that can ultimately promote the ERY degradation as shown in Fig. 10. The detail about ERY degradation via various photocatalysts is shown in Table 8.

8.2.2. Factors effecting photocatalysis

8.2.2.1. Catalyst concentration. In photocatalytic process the catalyst dosage is an important factor. As the reactive sites can be increased by increasing the catalyst dosage that can ultimately elevate the degradation rate. But the catalyst dosage is restricted to a limit. As the dosage crosses the limit the penetration of photons will be effected that decline the provided light energy through reflection and scattering. Moreover, studies reported that till certain increase of catalyst dosage the degradation mechanism increase due the availability of active sites but then eventually start decreasing as the number of active sites start declining. Xekoukoulotakis et al. (2010) concluded the increase of ERY with the increase in Znpc-TiO₂ catalyst concentration from 0.02 g/L to 0.5 g/L after 0.5 g/L the degradation rate started to decrease. Another experiment carried out with catalyst showed only 5% ERY degradation.

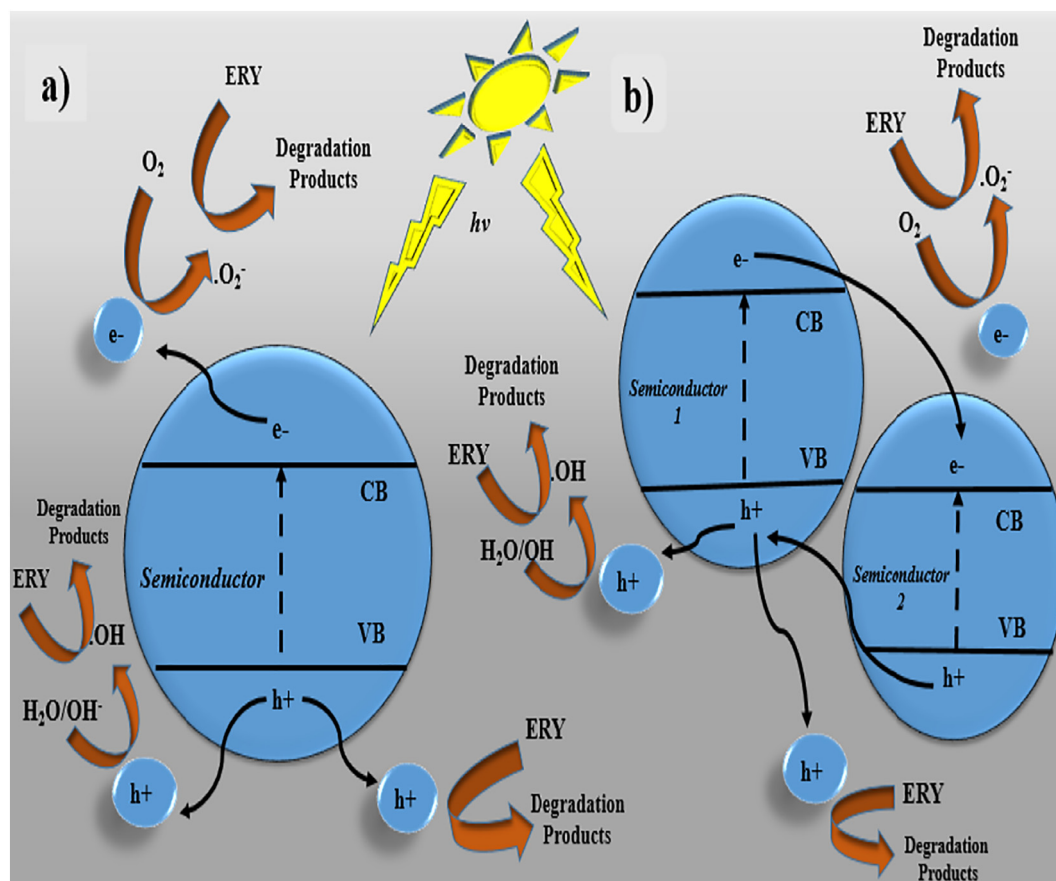


Fig. 9. Photocatalytic degradation of ERY (a) Single catalyst (b) nanocomposite.

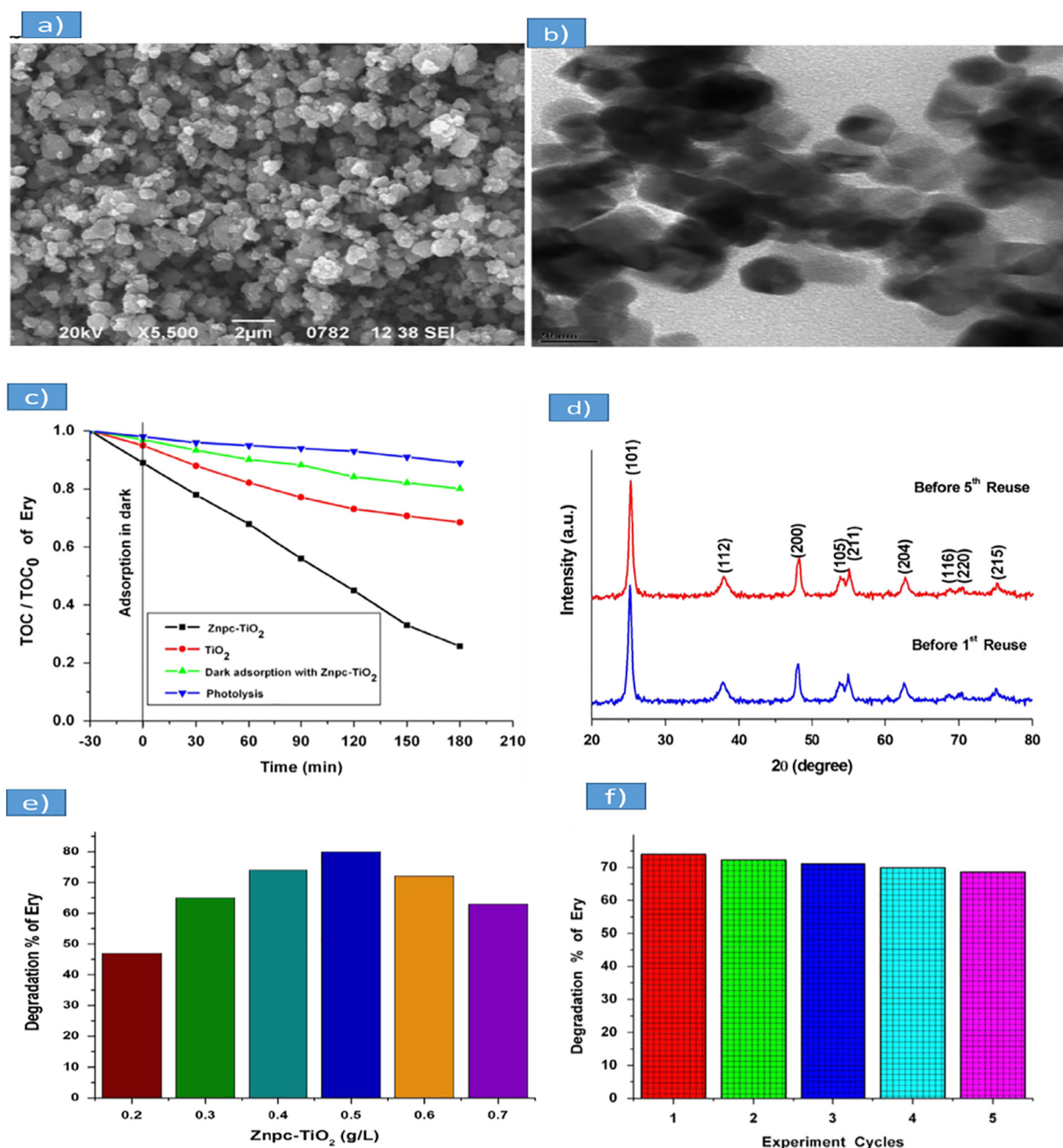


Fig. 10. a) SEM image of Znpc-TiO₂ and (b) TEM image of Znpc-TiO₂ (C) Photocatalytic degradation curves of ERY in the presence of TiO₂ and Znpc-TiO₂ at pH 5 under visible light irradiation (d) XRD patterns of Znpc-TiO₂ before and after the photoreaction (e) effect of Znpc-TiO₂ concentration on the photocatalytic degradation of ERY (f) Reusability of Znpc-TiO₂ for the photocatalytic degradation of ERY reprinted with permission of Elsevier (Vignesh et al., 2014a,b).

Table 8
Photocatalytic properties of ERY and its derivatives.

Contaminant	Photocatalyst	Catalyst concentration	Contaminant concentration	pH	Irradiation time	Irradiation source	Removal efficiency	References
ERY	ZnO/Fe ₃ O ₄	50–800 mg/L	10–100 μg/L	3–9	4 h	UVA	36%	(Fernández et al., 2019)
ERY	γ-Fe ₂ O ₃ /SiO ₂	500 mg/L	6 mg/L	8	6 min	UV	87.17%	(Fakhri et al., 2016)
AZY	GO@Fe ₃ O ₄ /ZnO/SnO ₂	1 g/L	30 mg/L	3	120 min	UV	90.06%	(Sayadi et al., 2019)
ERY	Znpc-TiO ₂	0.4 g/L	1 × 10 ⁵ M	5	180 min	UV	74.21%	(Vignesh et al., 2014a,b)
ERY	UV-A/TiO ₂	100–750 mg/L	2.5–30 mg/L	≤5	90 min	UV	90%	(Xekoukoulotakis et al., 2010)
ERY	TiO ₂ /carbon	1 g/L	10 mg dm ⁻³	–	300 min	UVA	87%	(Maletić et al., 2016)

AZY = azithromycin (ERY derivative).

However, highlighting the importance of catalyst in antibiotic degradation (Fakhri et al., 2016).

8.2.2.2. ERY concentration. The increase of ERY concentration decreases the degradation efficiency. This implies that ERY concentration was inversely proportional to its degradation efficiency keeping the catalyst dosage constant. A study confirmed that with the increase of ERY concentration from 2.5 to 30 mg/L keeping the catalyst dosage 500 mg/L the degradation efficiency decreased from 77% to 19% respectively (Xekoukoulotakis et al., 2010).

8.2.2.3. pH. For photocatalytic oxidation pH is termed out to be another important factor as it affects the process rate by governing water dissociation equilibrium, surface charge with respect to point of zero charge, ERY ionization and oxidative strength of light generated holes. Different charge form is shown by the substrate especially when pH is higher or lower than the substrates pKa (i.e. 8.8.) Similarly, the charge on the catalyst surface differs, when pH is greater or less as compared to that of the potential of zero charge of catalysts. Antibiotic removal capability is greater in basic pH as compared to that of the acidic one. Studies revealed a positive charge if the pH is <6.7 while the catalyst surface is negatively charged if the pH is >6.7. Acidic medium favors the photogenerated holes whereas alkaline or neutral medium favors the hydroxyl radicals. Therefore, mineralization of ERY is faster in low pH but for complete degradation alkaline medium is more favorable as ERY degradation. The most favorable ERY degradation pH range is 5–8 (Fakhri et al., 2016).

9. Conclusions and future perspectives

ERY being an abundantly used antibiotic is becoming a threat to the environment as its long term exposure can lead to the formulation of ERY-RBs and ERY-RGs. The food chain travelling property makes the picture worse as it becomes a threat to human health and ecosystem. The study records high concentrations of ERY in global water bodies. Conventional treatment technologies seem inefficient in treating ERY pollution. AOPs, because of their elevated degradation efficiency are considered to be effective and have shown remarkable results for ERY degradation. Out of three advance oxidation techniques discussed in this review photocatalysis is considered to be the most effective with least low economic value and high eco-friendliness for ERY degradation. In case of SO based oxidation processed no doubt the ERY degradation rate is high but that require high temperature, deal with harmful radiations, an activation source requirement make this process economically non feasible and it is ineffective for bulk degradations. Similarly, different studies suggest that ozonation is more effective towards the contaminants having aromatic structure as compared to macrolides. The slower rate of ERY degradation having entirely saturated structure is due to lack of one step pathway that makes the treatment process slow as well as O₃ sensitivity and energy consumption play their part in limiting the process efficiency. Progress has been witnessed in ERY degradation through AOPs but certain missing aspects are to be focused in future studies.

1. Optimization of AOPs for its practicability enhancement for instance use of eco-friendly catalyst.
2. Clarity in free radical generation mechanism and pollutant degradation mechanism is yet to be focused. Complete mechanism studies are still missing.
3. Further investigations regarding efficient removal of ERY-RGs and ERY-RBs are to be focused.
4. Complete mineralization of the antibiotic is not possible by single AOP. Therefore, integration of AOPs and other biological treatment processes can help overcoming this problem. By this way along with enhancing the efficiency the operational cost of the process can be reduced.

5. Along with antibiotics waste water is a combination of series of other organic and inorganic pollutants. The interaction between ERY and other pollutants can decrease the degradation efficiency therefore, there is a dire need of such studies instead of focusing on single antibiotic removal from aqueous medium. Attention to practical waste water is the requirement of the hour.
6. The application of AOPs on commercial scale is still missing. Future studies are required to address this area.
7. For starting any process analyzing the operational cost is the prior step. Studies regarding cost-effect analysis of AOPs should be conducted in future.
8. Health impacts due to the consumption of ERY concentrated water on human are still to be analyzed.

Declaration of competing interest

The authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

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